INFO248-project2ms1-Wong-Wu

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1 Overview

There are many factors that determine the salary of a job. There are aspects that the job requires and qualities that the applicant can uniquely bring. Knowing how each of these predictors affect the salary of a job is of major interest to employers and those looking to be employed as they help them gauge the market.

The main research question of this analysis is how do personal and job factors contribute to a the predicted salary of a job?

#Data Description and Processing The dataset used in this analysis was come from Kaggle[1]. We categorized the variables in the dataset as personal and job based. Knowing how these different factors affect possible job salaries is of major interest to both employers and applicants to gauge the market.

Personal Variables:
Experience years:Number of years of professional experience. Numeric
Education level:Highest level of education completed. Categorical
Skills count: Number of technical or professional skills. Numeric Certifications Number of professional certifications. Numeric

Job Variables:
Job title: The job role or position (e.g.Data Analyst, AI engineer). Categorical
Industry: Industry sector where the job belongs. Categorical
Company size: Size of the company (small, medium, large). Categorical
Location:Job location or region. Categorical
Remote Work:Whether the job allows remote work. Categorical

Outcome Variable: Salary: Annual salary of the employee. Numeric

The dataset was check for missing values. Of the original 250,000 observations, we found 0 observation that contained missing value.

2 Data Exploration

In figure 1, the distribution of the outcome variable, salary, shows a mostly symmetrical shape that is mildly skewed to the right.

Since the median is lower than the mean, the data is slightly right-skewed.

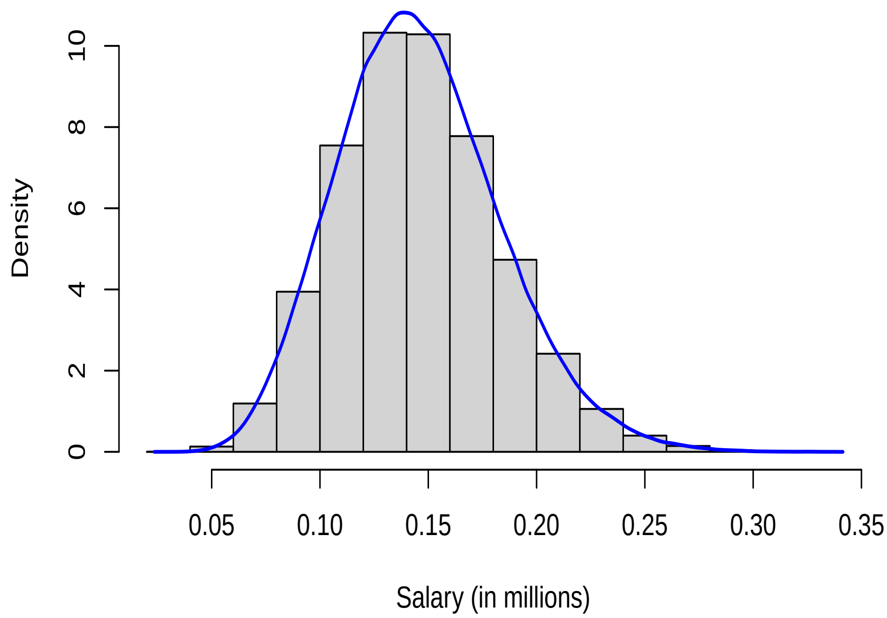

Figure 1: Salary distribution.

Table 1: Table 1) Salary descriptive statistics

statistic	value
Min	31867
Mean	145718.08
Median	143453
Max	333046
StdDev	37407.95

#Modeling

##Model 1

We fitted a linear regression model using all the predictors, with salary as the predicted variable.The estimates for the coefficients, standard errors and significance results are displayed below:

Table 2: Summary of coefficients for model 1.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	146739.297218	99.261495	1478.3103655	0.0000000
job_titleBackend Developer	-34409.353302	69.809940	-492.9004816	0.0000000
job_titleBusiness Analyst	-51570.078694	70.209192	-734.5203325	0.0000000
job_titleCloud Engineer	-21525.446790	70.176614	-306.7324782	0.0000000
job_titleCybersecurity Analyst	-24705.556779	69.945898	-353.2095184	0.0000000
job_titleData Analyst	-53786.744693	70.146657	-766.7755932	0.0000000
job_titleData Scientist	-26927.693748	70.004984	-384.6539518	0.0000000
job_titleDevOps Engineer	-23770.396116	70.005476	-339.5505254	0.0000000
job_titleFrontend Developer	-40914.297607	70.222215	-582.6403764	0.0000000
job_titleMachine Learning Engineer	-10727.889704	70.185487	-152.8505408	0.0000000
job_titleProduct Manager	-16166.146186	69.954565	-231.0949440	0.0000000

	Estimate	Std. Error	t value	Pr(>|t|)
job_titleSoftware Engineer	-32293.147714	70.015833	-461.2263584	0.0000000
experience_years	2697.076675	2.362575	1141.5836076	0.0000000
education_levelDiploma	-5330.889845	45.340910	-117.5735079	0.0000000
education_levelHigh School	-10712.143786	45.275905	-236.5970096	0.0000000
education_levelMaster	10808.343877	45.213557	239.0509510	0.0000000
education_levelPhD	21556.889935	45.323515	475.6226392	0.0000000
skills_count	857.951518	2.613253	328.3079160	0.0000000
industryEducation	75.798816	63.941847	1.1854336	0.2358472
industryFinance	55.119220	63.621947	0.8663554	0.3862961
industryGovernment	106.216410	63.934336	1.6613359	0.0966473
industryHealthcare	75.443079	63.936437	1.1799700	0.2380133
industryManufacturing	13.060797	63.857433	0.2045306	0.8379391
industryMedia	94.894863	63.848459	1.4862514	0.1372139
industryRetail	86.236130	63.973656	1.3479944	0.1776614
industryTechnology	60.042396	63.932378	0.9391547	0.3476522
industryTelecom	45.173266	63.961443	0.7062578	0.4800285
company_sizeLarge	-14095.834397	45.250956	-311.5035748	0.0000000
company_sizeMedium	-28270.787663	45.302021	-624.0513567	0.0000000
company_sizeSmall	-35381.225244	45.254660	-781.8250184	0.0000000
company_sizeStartup	-42367.764939	45.396113	-933.2905755	0.0000000
locationCanada	27947.671390	63.765383	438.2890872	0.0000000
locationGermany	14030.206276	64.022948	219.1433958	0.0000000
locationIndia	-41833.750183	63.938330	-654.2828068	0.0000000
locationNetherlands	-30.207911	63.960393	-0.4722909	0.6367196
locationRemote	-2.970812	63.829163	-0.0465432	0.9628774
locationSingapore	20.505700	63.847925	0.3211647	0.7480858
locationSweden	-26.311298	63.806197	-0.4123627	0.6800739
locationUK	20979.820861	63.917378	328.2334383	0.0000000
locationUSA	41940.355686	63.915126	656.1882669	0.0000000
remote_workNo	36.225400	35.029161	1.0341498	0.3010671
remote_workYes	5354.281032	35.103207	152.5296811	0.0000000
certifications	1615.901263	8.391013	192.5752188	0.0000000

#model 2 planning on using random forest, but not done

#References [1] https://www.kaggle.com/datasets/nalisha/job-salary-prediction-dataset